



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 6

Team No.: 16

Project Title: Alarm Clock

Team Members

Roll Number	Student Name	Signature
2520030537	Monisha Raghini Chelle	
2520030502	Shaik Sabiya Tabassum	
2520030453	Kolipaka Siri Chandana	

1. Abstract

The Linux Alarm Clock is a C-based, terminal-oriented user-space application designed to provide a simple and reliable alarm facility in Ubuntu/Linux. The project demonstrates Operating Systems and Systems Programming concepts through POSIX timing and signal mechanisms. The application will obtain the current system time, validate the user's alarm input, calculate the required waiting period, and schedule the alarm. When the timer expires, SIGALRM will notify the process and a configured signal handler will generate a clear terminal alert. Users will be able to set, view, and cancel an alarm through a simple menu-driven interface. The project will be implemented in C and compiled with GCC on Linux. Testing will cover valid and invalid time inputs, alarm setting, cancellation, triggering, and program termination. The expected outcome is a functional Linux-based Alarm Clock that can be compiled, executed, tested, and demonstrated while clearly illustrating practical use of Linux/POSIX system-programming mechanisms.

2. Problem Statement

A lightweight alarm application is required that can run directly in a Linux terminal and notify the user when a specified alarm time is reached, without depending on a graphical interface or external service. The project addresses this requirement while demonstrating how a Linux user-space process can use POSIX timing and signal mechanisms to wait for a timed event and respond to it. The application should provide clear alarm management, input validation, cancellation, and terminal output while keeping the implementation focused on Operating Systems and Systems Programming concepts.

3. Objectives

1. To develop a simple and reliable Alarm Clock application in C for Linux/Ubuntu.
2. To allow the user to set, view, and cancel an alarm through a terminal-based interface.
3. To use POSIX timing and signal mechanisms to schedule and handle the alarm event.
4. To test valid and invalid inputs and verify correct alarm triggering, cancellation, and termination.

4. Proposed Methodology

The project will be implemented as a C-based Linux user-space application. The program will display a menu and accept an alarm time from the user. The input will be validated and the required waiting duration will be calculated using the current system time. A Linux/POSIX alarm mechanism will then be configured for the calculated duration. When the timer expires, SIGALRM will be delivered to the process, and a signal handler configured using `sigaction()` will generate the alarm notification. The interface will also provide options to view the active alarm, cancel it, and exit. The application will be compiled with GCC on Ubuntu/Linux and verified using test cases for valid and invalid input, alarm triggering, cancellation, and program termination.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Signals – SIGALRM	Notifies the Alarm Clock process when the configured alarm timer expires.
<code>sigaction()</code>	Registers the POSIX signal handler used to handle SIGALRM and generate the alarm notification.
<code>alarm()</code>	Schedules a SIGALRM signal after the calculated waiting period.
<code>time()</code> / <code>localtime()</code>	Obtains and processes the current system time for alarm-time calculation and display.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030537	Monisha Raghini Chelle	Requirements analysis and project planning; preparation of the abstract and documentation; GitHub repository creation and maintenance; overall project coordination; integration support; final demonstration and viva preparation.
2520030502	Shaik Sabiya Tabassum	Support in implementation and testing; preparation of test cases and result verification; review of project documentation; assistance with screenshots and output evidence.
2520030453	Kolipaka Siri Chandana	Support in interface/flow design; assistance with implementation and debugging; preparation of presentation materials; support during project demonstration and viva.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development, compilation, execution, testing, and demonstration environment.
C	Implementation language for the Alarm Clock.
GCC	Compiles the C source code on Linux.
POSIX/Linux APIs	Provides signal and timing functionality for the alarm mechanism.
Git / GitHub	Version control and project repository for source code and documentation.

8. Expected Outcome

The project is expected to produce a working Linux-based Alarm Clock implemented in C. The application will allow the user to set, view, and cancel an alarm, wait for the configured time using a Linux/POSIX timing mechanism, and generate a clear terminal notification when the specified time is reached. The completed project will be tested with documented test cases and supported with output evidence, demonstrating both the functionality of the application and its practical use of Operating Systems and Systems Programming concepts.

9. GitHub Repository

GitHub Repository Link: https://github.com/ShaikSabiyaThabassum/2520030537_502_453_OS-SP_Project.git

Repository Created:

☒ Yes ☐ No

Faculty Name and Signature

Faculty Name: Kotte Anusha

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____